Course name:	Computer Graphics Fundamenals	**Course code**	CGD6214
Lecturer	Dr. Ng Kok Why	**Trimester**	2610
Programme	BCS	**Due Date**	Week 9 31st May 2026 (Sunday, 11.59pm)
Assignment	Assignment (Individual or max 2 persons)	**Weightage**	30%
Course outcome to achieve: Illustrate the basic concepts of 2D and 3D graphics.			

INSTRUCTION:

ASSIGNMENT: 3D Mini Driving Scene

1. Assignment Overview

Develop a small 3D driving simulation scene involving a moving vehicle, environment objects and animation.

2. Learning outcome:

By completing this assignment, students should be able to:

- Construct simple 3D geometric models procedurally
- Apply translation, rotation, and scaling transformations
- Implement hierarchical transformations
- Use view and projection matrices correctly
- Develop GLSL vertex and fragment shaders

3. Assignment Scenario:

Students are required to develop a vehicle model, road or track, buildings or obstacles to illustrate a graphical scene or environment. The scene should be rendered in real time using OpenGL and GLSL shaders.

4. Core Requirements:

a) The vehicle must contain body, wheels and windows. The wheel can rotate and the vehicle can translation.

b) For the environment, students shall create a ground plane and static scene objects. Move the vehicle along a path, rotate wheels and turn corner.

5. Required Controls

Involve keyboard navigation, mouse look or rotation, and zoom in/out for creating an interactive feeling.

Functional Features: Required User Interactions

Key	Function
W/A/S/D	Movement
+/-	Zoom in/out
Space	Pause animation
etc.	

6. Suggested Extensions (Bonus):

Students may attempt additional features for bonus marks:

- Texture mapping for the car / environment
- Skybox
- Headlights
- Particle stars for the Fog effects

7. Deliverables

Students must submit:

(i) A report to describe the assignment
(ii) Source code
(iii) Video demonstration (around 1 to 2 minutes).

In the report, kindly include below:

a) **A Cover Page**
   - Contain all the students' ID, names, Email address and phone number. (Refer to the sample in last page of this project write up.)

b) **Introduction**
   - Briefly introduce your project to make readers understand the main concept of the project.

c) **Documentations of virtual model**
   - Describe the purposes and its capabilities.
   - Explain how the objects are built, and how the animation you made works, if any.

d) **User Manual / Instructions**
   - Show how to interact with the program with keyboard and/or mouse, if any.

e) **Screen Shots**
   - A few pictures of the project you created.
   - Provide a short description for each picture.

Zipped the report, source code and video into a single file. Label the file as ***CGF_StudentName***.

Example for an individual: CGF_Mohamad_Ahmad_Zati
CGF_Jacky_Chan
CGF_Kanesaraj_Kumaran

Note: Just include 2 to 3 words for the name. Keep it short.

For those working in group, just include the first name from each student.

Example for 2 students in a group: CGF_Mohamad_and_Ali
CGF_Lee_and_Chan
CGF_Venushini_and_Kanes

Email the zip file to: **kwng@mmu.edu.my**

In your email, put the subject title as **CGF6214_Assignment**

It is your full responsibility to make sure that source code is error-free. Your code must be compatible with **Visual Studio 2022** or **2026** under Windows 10 or above. Do not include the executable file in the email!

END OF QUESTION

Assessment Rubric:

Component	**Marks**	**Actual Marks**
Modeling and Geometry	10	
Transformations / Animation	5	
GLSL Shaders	5	
User Interaction	3	
Code Structure	2	
Bonus: Additional features	5	
Total	30	

FACULTY OF COMPUTING AND INFORMATICS

CGD6214 COMPUTER GRAPHICS FUNDAMENTALS

ASSIGNMENT

Report

Lecture Section: TC1L

Lecturer Name: Dr. Ng Kok Why

Student ID	Student Name	Email Address	Phone No.